What is theory of automata?

Something that works automatically.

↳ It is something that deals with the study of abstract machines (mathematical models) as well as computational problems that can be solved using them.

What is mathematical about models?

↳ We have to prove:

1. Can the task be done at all?
2. When 1 is No, then this means this task cannot be done ever.

Machine and Associated languages

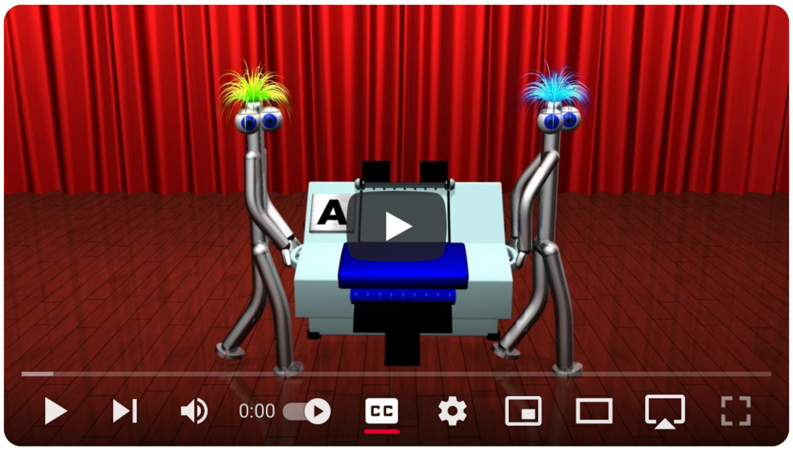
↳ When we introduce a new machine, we will try to learn it's language

↳ When we develop a language, we will try to find a machine that corresponds to it.

New Machines → New Languages: When we design new computational models, we explore the kinds of languages they can recognize or process.

New Languages → New Machines: When we define a new language or formalism, we try to build machines (like automata) to process or recognize it.

Proof machines can't do anything



Proof That Computers Can't Do Everything (The Halting Problem)

Languages:

↳ Considered as symbols with formal rules
↳ "Formal" emphasizes that it is form of the string of symbols that we are interested in, & not the meaning.

Alphabets:

↳ A finite non-empty set of symbols (called letters)
↳ Denoted by Σ
↳ Language is formed by concatenation of these letters
↳ Empty string is denoted by lambda Λ

Example,

$$\Sigma = \{x\}$$

$$L = \{x^n : n = 1, 2, 3, \dots\}$$

$$= \{x, xx, xxx, \dots\}$$

words of the language

How to identify valid strings/words?

↳ Tokenize the words into letters
↳ If the tokens exist in the alphabet set, only then it is valid

Example,

$$\Sigma = \{B, Ba, bab, d\}$$

String = BababB

(Ba)(bab)(B) → valid

(B)(abab)(B) → Invalid

↳ Reverse of string is denoted by Rev(s) or reverse(s) or s^r.

Palindrome:

$$rev(s) = s$$

Defining a Language:

↳ Either tell us how to test if a string is valid or not OR tell us the how to construct the words in a language by a rule.

Example,

$\Sigma = \{a\}$

Rule: Has odd length

Kleene Closure: → zero or more

↳ Given Σ, define a language in which any string of letters from Σ of all lens is a word including Λ.

↳ It is denoted by Σ^*

Example,

Consider $\Sigma = \{x\}$

$\Sigma^* = \{\Lambda, x, xx, xxx, \ldots\}$ → infinite words each of finite length

↳ This is in Lexiographic Order. From shortest to largest

Plus Closure: → Mandatory when on a letter

↳ Given Σ, define a language in which any string of letters from Σ of all lens is a word.

↳ It is denoted by Σ^*

Example,

Consider $\Sigma = \{x\}$

$\Sigma^* = \{x, xx, xxx, \ldots\}$ → infinite words each of finite length

↳ This is in Lexiographic Order. From shortest to largest